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21 DAYS SQL CHALLENGE

CHALLENGE STARTS FROM

3RD NOVEMBER 2025

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#SQLWithIDC

Day 19 (25/11): Window Functions - ROW_NUMBER, RANK, DENSE_RANK

🎯 Objective

To understand and apply SQL window functions for ranking and ordering rows within partitions.
By the end of this lesson, you will be able to:

- Assign unique row numbers within groups
- Generate rankings with or without gaps
- Compare and analyze records within categories using window functions

🔍 Topics Covered

- ROW_NUMBER() - unique sequence number
- RANK() - ranking with gaps
- DENSE_RANK() - ranking without gaps
- PARTITION BY - group-wise ranking

★ *What is a Window Function?*

A window function helps you look at one row while also looking at other rows around it.

It does not hide or group the data — it keeps all rows as they are.

☞ *Think of it like this:*

You are standing in a classroom.

You can see **your own marks AND your classmates' marks** at the same time.

A window function works the same way:

- It looks at the current row
- And also looks at a “window” of other rows (classmates)

ROW_NUMBER() - Unique Sequence Number

Gives a unique number to each row inside a group.
No duplicates, no ties.

Syntax

```
ROW_NUMBER() OVER(PARTITION BY column ORDER BY column)
```

Example

List employees with a unique number inside each department.

```
SELECT
  employee_name,
  department,
  ROW_NUMBER() OVER (
    PARTITION BY
    department
    ORDER BY salary DESC
  ) AS row_num
FROM employees;
```

Within each department, highest salary gets row_num = 1, next = 2, and so on.

RANK() - Ranking With Gaps

Same values get the same rank.
But next rank jumps (gaps).

Syntax

```
RANK() OVER(PARTITION BY column ORDER BY column)
```

Example

Rank students by marks inside each class..

```
SELECT
  student_name,
  class,
  marks,
  RANK() OVER (
    PARTITION BY class
    ORDER BY marks DESC
  ) AS rank_num
FROM students;
```

If two students have same marks → same rank.
Next rank will skip (1, 2, 2, 4).

DENSE_RANK() - Ranking Without Gaps

Same values have same rank, but **no skipping**.

Syntax

```
DENSE_RANK() OVER(PARTITION BY column ORDER BY column)
```

Example

Rank students by marks inside each class..

```
SELECT
  student_name,
  class,
  marks,
  DENSE_RANK() OVER (
    PARTITION BY class
    ORDER BY marks DESC
  ) AS dense_rank_num
FROM students;
```

If marks are tied → same rank.
Next rank continues (1, 2, 2, 3).

PARTITION BY - Group-wise Ranking

It divides your data into groups.
Ranking starts fresh **inside each group**.

Example

```
ROW_NUMBER() OVER(PARTITION BY department ORDER BY salary DESC)
```

PARTITION BY department → Separate ranking for each department
No partition = ranking for the whole table

Practice Questions:

```
-- Rank patients by satisfaction score within each service.
SELECT
  patient_id,
  name AS patient_name,
  satisfaction,
  RANK() OVER (
    PARTITION BY service
    ORDER BY satisfaction DESC
  ) AS satisfaction_score_num
FROM patients;
```

patient_id	patient_name	satisfaction	satisfaction_score_num
PAT-bdf625cd	Brent Hernandez	99	1
PAT-8bfedd28	Amanda Harvey	99	1
PAT-c3e8e421	Holly Valentine	99	1
PAT-581a47d8	Paul Castaneda	99	1
PAT-87fa07bd	Martha Smith	99	1
PAT-8f6d5fa4	Bryan Herrera	99	1
PAT-87783e44	Devon Flores	99	1
PAT-f9093e10	Rebecca Kelly	99	1
PAT-27519dd1	David Harris	98	9
PAT-3e5b33b6	Kim Erickson	98	9
PAT-f801ed9b	Dawn Mullins	98	9
PAT-40efa3f6	Zachary Robinson	98	9
PAT-d70a9eaa	Susan Murray MD	98	9
PAT-14171b1d	Gary Wells	97	14
PAT-8a8a2306	Thomas Peterson	97	14
PAT-553df120	James Watson	97	14
PAT-f58ab9a1	Megan Green	97	14
PAT-71583d93	Kiara McIntyre	97	14
PAT-f8e37525	Thomas Harris	97	14
PAT-44c855d4	Stephanie Salazar	97	14
PAT-0b1acdd9	Aaron Schroeder	97	14

Practice Questions:

```
-- Assign row numbers to staff ordered by their name.  
SELECT  
    staff_id,  
    staff_name,  
    ROW_NUMBER() OVER (  
        ORDER BY staff_name ASC  
    ) AS row_num  
FROM staff;
```

staff_id	staff_name	row_num
STF-1881ede3	Aaron Wise	1
STF-39082289	Abigail Shaffer	2
STF-3c9039fc	Aimee Montoya	3
STF-5ca26577	Allison Hill	4
STF-48fc5275	Amber Kidd	5
STF-b5fcdde5	Amy Silva	6
STF-9a4c0716	Angela Lopez	7
STF-177e5df8	Angelica Tucker	8
STF-d8006e7c	Angie Henderson	9
STF-e328da72	Anna Henderson	10
STF-36b3e876	Anthony Hump...	11
STF-15c07995	Anthony Rodrig...	12
STF-6a7114cb	April Frost	13
STF-4740993e	Ashley Waller	14
STF-bcb3f4fe	Brent Jordan	15
STF-768c9d6f	Brian Ramirez	16
STF-446cb017	Brian Rodriguez	17
STF-3007d318	Brittany Farmer	18
STF-a74d38db	Brittney Phillips	19
STF-fdc1ae8e	Cameron Parker	20
STF-434f45a6	Carla Gray	21

Practice Questions:

```
-- Rank services by total patients admitted.
SELECT
    service,
    total_patients_admitted,
    DENSE_RANK() OVER (
        ORDER BY total_patients_admitted DESC
    ) AS dense_rank_num
FROM (
    SELECT
        service,
        SUM(patients_admitted) AS total_patients_admitted
    FROM services_weekly
    GROUP BY service
) AS t;
```

service	total_patients_admitted	dense_rank_num
general_medicine	2332	1
surgery	1686	2
emergency	1185	3
ICU	648	4

Daily Challenge:

```
-- For each service, rank the weeks by patient satisfaction score (highest first).
-- Show service, week, patient_satisfaction, patients_admitted, and the rank.
-- Include only the top 3 weeks per service.
SELECT *
FROM (
    SELECT
        service,
        week,
        patient_satisfaction,
        patients_admitted,
        RANK() OVER (
            PARTITION BY service
            ORDER BY patient_satisfaction DESC
        ) AS satisfaction_rank
    FROM services_weekly
) AS ranked_data
WHERE satisfaction_rank <= 3;
```

service	week	patient_satisfaction	patients_admitted	satisfaction_rank
emergency	19	99	16	1
emergency	48	99	20	1
emergency	11	98	16	3
general_medicine	45	99	40	1
general_medicine	1	97	37	2
general_medicine	47	97	55	2
ICU	16	98	15	1
ICU	26	98	8	1
ICU	48	98	16	1
surgery	47	99	48	1
surgery	20	99	31	1
surgery	36	97	47	3
surgery	14	97	33	3